

FAQ:

TL;DR check out or post on <https://github.com/brainchart/Lifespan/>, file any issues there or send an email to BrainCharts@gmail.com with any questions.

- Where can I get more information or help with specific questions pertaining to my dataset?
 - Check the readme and make sure that your data is similar to the template
 - Check out <https://github.com/brainchart/Lifespan/> or email BrainCharts@gmail.com
- Where can I get the centile scores for publicly-available datasets that are already included in the lifespan models?
 - Reach out directly with verification of approval to access the given dataset(s), and we can securely share the de-identified centile scores. We ask that you cite our paper (<https://www.nature.com/articles/s41586-022-04554-y>) if using these scores in published materials.
- Can I/we use this tool to generate individual centile/percentile scores in our clinic/centre/study?
 - The online tool provides the opportunity to generate centile scores for dataset outside of the original model (and for those within), and, for the most robust centile scores, we recommend a minimum sample size of at least 100 individuals without a diagnosis acquired with the same protocol to generate a proper site-/study-specific parameterisation. It is not (yet) suited to generate centile/percentile scores for individual scans.
- Can I get confidence intervals on my (newly generated) centile scores?
 - Currently the bootstrapping procedure required to generate individual level confidence intervals (such as those shown in the manuscript) are too computationally intensive to be run through our online interface without significantly restricting access. Depending on our availability we can assist with validation and bootstrapping of new data to generate individual confidence intervals offline and should this be of interest we do encourage people to reach out or post on our GitHub page.
- Can I use this tool for both cross-sectional and longitudinal datasets?
 - Yes! Our models were constructed using cross-sectional data (one time point per individual), but validated against longitudinal data which was available for a subset of the overall sample. The out-of-sample estimator will therefore generate the study-specific parameterization using just the first time point (indicated as “session”==1). We recommend proceeding with caution and reaching out directly, especially for datasets with uneven distributions of longitudinal data (staggered timepoints, or sparse sampling) or for longitudinal datasets spanning periods of high developmental change (e.g., perinatal, infancy, early childhood). We are in the process of further developing and validating our methods for these types of datasets.

- Should incorporate gestational age information when available?
 - Yes, when available we incorporated this information when calculating age in days. For datasets that don't have this information, please add 280 to the age in days for each subject (arbitrarily set at full term=40 weeks=280 days). This is particularly important for datasets with younger subjects when the curves are most dynamic, since omitting this information will cause the resultant centile scores to be heavily skewed.
- The app doesn't work for me. I can see the top plot but not the centiles?
 - In most cases this is likely caused by a column name mismatch between what the app expects and the data provided. First please check our template to make sure these match. Second, if your (multi-site) dataset includes either no control subjects at each site or does not contain at least 10 control subjects per sex per site the app will not compute centiles.
- Can I access the raw data? Is it openly available?
 - Unfortunately due to various agreements signed as part of this project and the fact that our database includes unpublished proprietary imaging datasets we are not able to share all data openly. The website and paper (first table in the Supplemental Information) contain references and links to all datasets included where you could obtain or request access to the listed datasets. Strictly for non-commercial use all models are made available through GitHub alongside the code-base and tutorials.
- Can I contribute data?
 - We are always keen to improve and update the models, especially to increase the representativeness, and to include contributors as members of the consortium in future work. Do reach out to us directly with a very brief description of your dataset, and we will respond as soon as possible.
- I have run the app and see the centiles, but when I download my centiles there are only centiles for a single phenotype?
 - We currently have the app setup to compute centiles when a certain feature/phenotype is selected to avoid unnecessarily using resources (i.e., not every centile feature might be needed) to compute centiles that are not needed. Selecting a different feature/phenotype and then downloading the output again will include only the selected phenotype.
- If I have multi-site data does the app harmonise this automatically?
 - The app currently runs site-specific GAMLSS models, however as we note in our accompanying manuscript this harmonisation is optimal mainly for sites that have at least 100 or more subjects per site. Depending on your situation it may be appropriate to assess site effects first and possibly harmonise data prior to generating centiles on brainchart.io (for example by using an appropriate ComBAT harmonisation strategy first to account for study-specific factors that are not included in our lifespan models).
- Will the models be updated regularly?

- We are continuously receiving more data to update our models and given the computational burden to generate models and validate them we plan to update them at least every 6 months.
- Does the app support other brain imaging features?
 - We are always working on expanding the set of available features but have no fixed timeline on when these might be available. Reach out directly if you are interested in specific features that aren't yet included!
- I have an idea about how to use this for a new question/project, can I use your data or do you want to collaborate on it?
 - We are always open to collaborate though we should note that our time and resources are obviously limited. We strongly encourage you to reach out directly about this and to discuss possibilities and/or make connections to other groups in the consortium that may be able to help or collaborate.